

TOBOLSK, Russian Federation

WMO: 282750

Lat: 58.150N

Lon: 68.250E

Elev: 154

StdP: 14.61

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-26.7	-36.6	0.7	-31.6	-31.6	0.9	-26.5	18.3	13.9	16.6	10.5	2.4	320	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	83.8	67.6	80.9	66.6	77.8	64.8	70.1	80.4	68.3	77.6	66.5	75.2	6.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.9	75.3	64.7	92.1	73.3	63.0	86.5	71.6	34.2	80.7	32.6	77.7	31.3	75.3	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.0	14.7	13.3	DB	-38.2	89.1	6.3	1.8	-42.7	90.4	-46.4	91.5	-50.0	92.5	-54.6	93.8	
			WB	-38.3	72.6	6.3	1.4	-42.8	73.6	-46.5	74.4	-50.0	75.2	-54.6	76.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.9	-0.7	4.3	20.1	36.3	50.7	61.7	65.2	59.4	48.9	37.0	17.0	5.1
	DBStd	25.42	14.49	13.82	10.82	10.46	9.70	8.11	6.50	6.94	7.63	8.88	14.17	15.55
	HDD50	7236	1572	1281	927	422	114	9	0	6	111	411	991	1392
	HDD65	11531	2037	1701	1392	862	449	156	82	201	484	869	1441	1857
	CDD50	1360	0	0	0	10	137	360	470	298	78	7	0	0
	CDD65	180	0	0	0	0	7	58	86	28	1	0	0	0
	CDH74	1585	0	0	0	2	130	520	683	234	16	0	0	0
	CDH80	334	0	0	0	0	18	130	142	43	1	0	0	0
Wind	WSAvg	6.0	5.9	5.8	7.0	7.0	6.9	5.7	4.9	4.8	5.4	6.0	6.2	6.0
Precipitation	PrecAvg	18.7	0.9	0.6	0.7	0.9	1.7	2.6	2.6	2.9	1.9	1.6	1.4	1.1
	PrecMax	29.4	1.6	1.3	2.2	2.5	4.1	6.4	9.2	7.7	3.8	4.0	3.0	2.6
	PrecMin	13.1	0.1	0.0	0.0	0.0	0.2	0.6	0.3	0.2	0.3	0.0	0.1	0.0
	PrecStd	2.9	0.4	0.3	0.5	0.6	0.8	1.4	1.4	1.7	0.9	0.9	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	37.4	45.6	71.1	83.0	87.8	87.1	84.6	76.0	64.8	44.4	36.3
		MCWB	32.7	33.2	38.0	54.9	60.6	68.5	70.6	69.0	60.3	51.6	40.2	33.6
	2%	DB	29.3	33.5	40.7	63.0	77.9	83.6	83.8	80.2	70.5	58.8	39.5	33.0
		MCWB	27.8	30.9	34.8	51.1	59.4	67.3	68.7	67.2	58.8	49.1	37.1	31.4
	5%	DB	24.8	29.2	37.6	56.8	73.7	80.1	81.0	76.2	65.5	53.6	36.4	29.4
		MCWB	23.3	26.9	32.8	46.8	57.6	65.8	67.7	65.1	56.1	46.8	34.5	27.8
	10%	DB	20.2	24.7	34.9	51.9	68.7	76.3	77.9	72.0	61.0	49.7	33.7	25.6
		MCWB	18.9	22.5	31.3	43.7	54.9	64.2	65.8	62.8	53.8	44.6	32.1	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.7	33.8	38.7	56.0	63.7	71.9	72.7	71.2	62.5	53.1	40.6	34.3
		MCDB	34.5	36.1	44.2	67.3	77.5	83.9	83.8	82.0	72.4	60.9	43.1	35.8
	2%	WB	27.9	31.1	35.4	51.5	60.9	69.1	70.7	68.2	59.5	50.4	37.3	31.6
		MCDB	29.2	33.7	39.6	62.0	75.2	80.1	80.9	77.4	68.3	56.9	39.0	33.3
	5%	WB	23.3	27.1	33.5	47.6	58.4	67.2	69.0	66.1	57.0	47.7	34.6	27.9
		MCDB	24.6	29.1	36.7	55.9	71.1	77.6	78.1	74.3	63.9	53.0	36.2	29.3
	10%	WB	18.9	22.7	31.5	43.9	55.9	65.0	67.3	63.9	54.5	45.0	32.3	24.2
		MCDB	20.2	24.6	34.8	51.1	67.3	74.4	75.5	70.4	60.2	49.4	33.7	25.6
Mean Daily Temperature Range	5% DB	MDBR	14.4	18.1	18.4	17.7	19.7	17.5	17.0	16.4	15.0	12.1	11.6	13.7
		MCDBR	14.1	15.5	17.4	25.5	27.7	24.0	21.9	23.0	22.6	20.1	11.1	11.9
		MCWBR	13.6	13.8	12.9	15.3	13.4	10.5	9.4	11.1	12.0	12.0	9.7	11.4
	5% WB	MCDBR	13.7	15.0	15.0	24.6	25.1	21.7	19.6	21.0	20.9	19.1	10.6	11.9
		MCWBR	13.3	13.6	12.0	15.2	13.3	10.2	9.2	10.8	12.0	11.9	9.7	11.5
Clear-Sky Solar Irradiance	taub	0.278	0.312	0.371	0.410	0.404	0.408	0.417	0.426	0.377	0.346	0.304	0.275	
	taud	2.016	2.014	1.931	2.054	2.255	2.308	2.288	2.223	2.324	2.254	2.089	1.969	
	Ebn at Noon	165	217	234	249	263	263	257	244	239	207	153	122	
	Edh at Noon	18	29	42	44	39	38	37	37	29	23	17	14	
All-Sky Solar Radiation	RadAvg	200	504	970	1379	1629	1775	1688	1276	808	401	177	114	
	RadStd	16	38	65	129	120	126	161	111	85	52	33	15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page